

Esercizi

Edoardo Longo

1. *Svolgere i seguenti esercizi:*

- (a) Sapendo che $\cos \alpha = -3/5$ e $\sin \beta = 2/3$ con $\pi < \alpha < 3\pi/2, \pi/2 < \beta < \pi$ calcola:
 $\sin(\alpha + \beta), \cos(\alpha + \beta), \sin(\alpha - \beta), \cos(\alpha - \beta)$.
- (b) Sapendo che $\sin \alpha = 1/3$ con $\pi/2 < \alpha < \pi$ calcola: $\sin(2\alpha), \cos(2\alpha), \tan(2\alpha)$.
- (c) Sapendo che $\alpha = \arctan 3$ calcola: $\sin(2\alpha), \cos(2\alpha), \tan(2\alpha)$.

2. *Risolvere le seguenti equazioni goniometriche elementari:*

- (a) $\sin x = 1$
- (b) $5 \sin x = 6$
- (c) $\sin x = -\frac{1}{4}$
- (d) $2 \cos x = \sqrt{3} \tan\left(-\frac{\pi}{4}\right)$
- (e) $\frac{9}{2} \sin x = -4\sqrt{3} - \frac{7}{2} \sin x$
- (f) $\sin x = \sin \frac{\pi}{6} + \cos \frac{\pi}{3}$
- (g) $\tan x = \sin\left(-\frac{\pi}{6}\right) + \cos \frac{2\pi}{3}$
- (h) $\sin\left(\frac{\pi}{2} - x\right) = \frac{1}{2} \tan\left(\frac{\pi}{3}\right)$
- (i) $\frac{3}{2} \cos x + \sin \frac{\pi}{2} = \cos \frac{\pi}{3}$

3. *Risolvere le seguenti equazioni goniometriche:*

- (a) $2 \sin 3x = 1$
- (b) $\sin\left(3x - \frac{\pi}{3}\right) = -1$
- (c) $\sin\left(x + \frac{\pi}{6}\right) = \tan \pi$
- (d) $-\cos 10x + \frac{1}{2} = \frac{3}{2} - 2 \cos 10x$
- (e) $3 \tan\left(3x - \frac{\pi}{6}\right) = \sqrt{3}$
- (f) $\frac{\pi}{5} \cos\left(x - \frac{\pi}{6}\right) = 0$
- (g) $\frac{5}{3} - \tan\left(2x + \frac{\pi}{6}\right) = \frac{8}{3}$
- (h) $\cos\left(2x + \frac{\pi}{3}\right) = -\frac{1}{2}$
- (i) $\tan\left(x + \frac{2\pi}{3}\right) = \sqrt{3}$

4. *Risolvere le seguenti equazioni goniometriche:*

- (a) $2 \sin 3x = 1$
- (b) $\sin x = \sin\left(\frac{\pi}{2} - 3x\right)$
- (c) $\cos\left(2x - \frac{\pi}{2}\right) = \cos\left(x + \frac{\pi}{4}\right)$
- (d) $\cos\left(\frac{\pi}{2} - x\right) = \cos\left(x - \frac{\pi}{3}\right)$
- (e) $\tan\left(x + \frac{\pi}{3}\right) = \tan\left(2x - \frac{\pi}{4}\right)$
- (f) $\sin\left(\frac{\pi}{2} - \frac{x}{2}\right) = \cos\left(\frac{\pi}{5} - x\right)$
- (g) $\sin\left(x + \frac{\pi}{3}\right) = -\cos x$